Intelligent 3DCP: Mechanical Strength Prediction Using Physics-Informed Masked Multi-Task Learning

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***Abstract*—Three-Dimensional Concrete Printing (3DCP) requires reliable and non-destructive estimates of hardened concrete strength. This study considers three related properties: compressive strength (C_s), splitting tensile strength (T_s), and flexural strength (F_s). In the ConcreteXAI database of 4,420 specimens, compressive strength is available for all samples, whereas tensile and flexural measurements are missing for 21.7% and 60.2% of samples, respectively. Consequently, only 39.8% of specimens contain measurements for all three properties. Removing incomplete samples reduces the usable data, while imputing missing values can introduce errors into the learning process. Standard neural networks may also produce physically implausible predictions, such as tensile strength exceeding compressive strength ($T_s \geq C_s$).**

This paper proposes Physics-Informed Masked Multi-Task Learning (PIM-MTL), which learns shared representations from all 4,420 specimens while masking the loss gradients for unobserved targets. A differentiable quadratic penalty further encourages predictions to remain within the specified strength-ratio bounds: $0.06 \leq T_s/C_s \leq 0.18$ and $0.08 \leq F_s/C_s \leq 0.25$. On a held-out test set of 884 specimens, PIM-MTL obtains R^2 values of 0.9818, 0.9857, and 0.9972 for compressive, splitting tensile, and flexural strength, respectively. The corresponding RMSE values are 1.3143, 0.1097, and 0.1262 MPa. The model also records a 0.00% test-set violation rate for $T_s \geq C_s$. These results indicate that masked multi-task learning with physics-based regularization can use incomplete experimental data while supporting physically consistent strength prediction for 3DCP.

***Index Terms*—3D Concrete Printing, Mechanical Strength Prediction, Multi-Task Learning, Physics-Informed Neural Networks, Asymmetric Missingness, Masked Gradient Gating, ACI 318 Compliance.**

I. Introduction

Automated construction and Three-Dimensional Concrete Printing (3DCP) represent a paradigm shift in civil and architectural engineering, offering unprecedented geometric freedom, material optimization, and substantial reductions in labor costs and construction timelines [1], [2]. Unlike traditional mold-cast concrete, 3D-printed elements rely upon layer-by-layer extrusion without temporary formwork. Consequently, real-time assessment and predictive modeling of mechanical strength are vital to verify structural integrity, buildability, and long-term durability [3].

In structural concrete design (governed by standards such as ACI 318 [4] and Eurocode 2 [5]), structural performance cannot be characterized by compressive resistance alone. Instead, safety and serviceability require knowledge of three coupled mechanical strengths:

1) **Compressive Strength (C_s, MPa):** Uniaxial crushing resistance (ASTM C39 [6]), defining fundamental axial load-bearing capacity.
2) **Splitting Tensile Strength (T_s, MPa):** Diametral tensile capacity (ASTM C496 [7]), critical for predicting crack initiation, diagonal tension, and shear capacity.
3) **Flexural Strength / Modulus of Rupture (F_s, MPa):** Extreme fiber bending capacity (ASTM C78 [8]), governing slab deflection, cracking moments, and pavement durability.

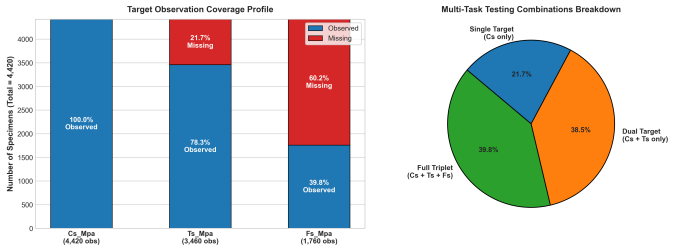


Fig. 1. Target observation and asymmetric missingness profile across the 4,420 experimental mixes in the ConcreteXAI benchmark dataset.

Despite their shared dependence on cement hydration kinetics, water-to-binder ratio, aggregate packing, and curing maturity, gathering concurrent experimental measurements for all three parameters is exceedingly difficult. Compressive cylinder crushing is fast and standardized. Conversely, splitting tensile and third-point flexural testing require specialized test fixtures, heavy prism specimens ($150 \times 150 \times 500$ mm), and extensive manual labor.

Consequently, real-world multivariate concrete databases exhibit severe *asymmetric missingness*. In the international ConcreteXAI benchmark database [9] comprising 4,420 multivariate specimens, compressive strength is 100% complete, but splitting tensile strength is missing in 21.7% of specimens ($N = 960$), and flexural strength is missing in 60.2% ($N = 2,660$). As shown in Fig. 1, only 1,760 out of 4,420 mixes (39.8%) possess complete three-property labels.

This research was conducted in the Department of Information Technology at National Institute of Technology Karnataka (NITK), Surathkal, India.

Standard machine learning practices fail in this setting:

- **Complete-Case Analysis (Row Dropping):** Discarding incomplete records discards 60.2% (2,660 mixes) of experimental data, destroying statistical power and inducing severe laboratory selection bias.
- **Artificial Imputation:** Filling unobserved strengths with mean, median, or k-nearest neighbor values introduces synthetic artifacts that corrupt empirical gradients.
- **Independent Single-Task Regressors:** Training separate models for each property ignores the continuum mechanics coupling between compression, tension, and rupture.
- **Unconstrained Neural Networks:** Standard multi-task networks frequently produce physically absurd outputs, predicting negative strengths, tensile capacities exceeding compressive strength ($T_s \geq C_s$), or out-of-code ratios.

To overcome these fundamental limitations, this work develops a **Physics-Informed Masked Multi-Task Learning (PIM-MTL)** architecture. The primary contributions of this paper are:

1) **Masked Multi-Task Gradient Gating:** A unified multi-task loss formulation that enables training on all 4,420 mixes simultaneously. A binary observation mask ensures that unobserved targets receive strictly zero backpropagation gradient, utilizing 100% of the data without synthetic imputation.
2) **Differentiable Continuum Physics Regularization:** Embedding ACI 318 and Eurocode 2 ratio bounds ($0.06 \leq T_s/C_s \leq 0.18$ and $0.08 \leq F_s/C_s \leq 0.25$) directly into the PyTorch optimization graph via a two-sided quadratic hinge penalty evaluated on all specimens, regularizing unobserved heads via microstructural mechanics.
3) **Production-Grade Deployment Architecture:** Incorporating a custom `SafeBatchNorm1d` layer that eliminates runtime exceptions during single-specimen ($N = 1$) on-site inference, combined with Gaussian Error Linear Units (GELU) and Cosine Annealing optimization.
4) **Rigorous Quantitative and Physical Validation:** Demonstrating state-of-the-art predictive performance ($R^2 > 0.98$ on all targets) and verifying a 0.00% rate of physically impossible predictions ($T_s \geq C_s$) across an isolated 884-sample test split.

II. Related Work

A. Machine Learning in Concrete and 3DCP

Recent literature has increasingly explored data-driven methods for concrete property prediction. Ali et al. [2] investigated Support Vector Machines (SVM) and XGBoost regression to predict the tensile and flexural strength of 3D-printed concrete from mix formulations. Zhang et al. [3] examined Random Forests and Artificial Neural Networks (ANN) for mechanical strength estimation, noting strong correlations with binder composition. Similarly, Feng et al. [10] synthesized an extensive international database to assess compressive capacity. However, these prior studies treated strength prediction as isolated single-target regression tasks on complete datasets, discarding unobserved targets and neglecting multi-property mechanics.

B. Multi-Task Learning and Incomplete Supervision

Multi-Task Learning (MTL) [11] aims to improve generalization by sharing representations across related tasks. In concrete mechanics, compressive, tensile, and flexural strengths originate from the same microstructural hydration products (calcium-silicate-hydrate gel, pore structure, and interfacial transition zone). Hence, an encoder trained on multiple targets should theoretically discover richer representations than disjoint models. However, classical MTL architectures assume complete target matrices. In sparse engineering datasets, naive masking often leads to gradient imbalance or vanishing updates. Developing exact gradient-gating mechanisms is crucial to maintain representation learning without distorting loss dynamics.

C. Physics-Informed Neural Networks (PINN)

Physics-informed machine learning, formalized by Karniadakis et al. [12], integrates governing physical laws (such as partial differential equations, conservation principles, and empirical bounds) into deep neural loss functions. In structural concrete research, Feng et al. [10] demonstrated that penalizing unphysical strength behaviors improves out-of-distribution robustness. In this paper, we extend physics regularization to multi-task learning under extreme missingness: by evaluating continuum ratio constraints on *all* samples, the physical regularizer provides meaningful gradient guidance to unobserved prediction heads even in the total absence of ground-truth laboratory measurements.

III. Problem Formulation and Mechanical Constraints

A. Mathematical Problem Setting

Let $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i, \mathbf{m}_i)\}_{i=1}^{N}$ denote a multivariate concrete dataset of $N = 4,420$ specimens. Each input vector $\mathbf{x}_i \in \mathbb{R}^D$ comprises material composition, non-destructive test measurements, and curing age. The target vector is defined as:

$$\mathbf{y}_i = \begin{bmatrix} C_{s,i} & T_{s,i} & F_{s,i} \end{bmatrix}^T \in \mathbb{R}^3 \tag{1}$$

where $C_{s,i}$, $T_{s,i}$, and $F_{s,i}$ represent compressive, splitting tensile, and flexural strengths in Megapascals (MPa).

Because experimental observations are incomplete, each sample is associated with a binary observation mask:

$$\mathbf{m}_i = \begin{bmatrix} m_{i,1} & m_{i,2} & m_{i,3} \end{bmatrix}^T \in \{0, 1\}^3 \tag{2}$$

where $m_{i,j} = 1$ if target j was experimentally measured for specimen i, and $m_{i,j} = 0$ if the target is unobserved. For unobserved entries, a computational placeholder $y_{i,j} = 0.0$ is assigned in memory. The objective is to learn a parameter-shared neural network $\hat{\mathbf{y}}_i = f_\theta(\mathbf{x}_i)$ that accurately predicts all three mechanical properties while strictly respecting physical continuum bounds.

B. Continuum Mechanics and Civil Engineering Code Envelopes

Concrete is a quasi-brittle composite material whose tensile fracture is governed by microcrack propagation through the Interfacial Transition Zone (ITZ). Over a century of empirical testing and fracture mechanics formulations codified in ACI 318-19 [4] and Eurocode 2 [5] define strict physical relationships between compressive, tensile, and flexural strengths.

1) Splitting Tensile to Compressive Ratio (R_t): According to ACI 318 Section 19.2.4 and Eurocode 2 Section 3.1.2, direct and splitting tensile capacities scale with the square root of compressive strength:

$$f_{ct} \approx 0.53\sqrt{f'_c} \quad \text{to} \quad 0.59\sqrt{f'_c} \quad \text{(MPa)} \tag{3}$$

For typical concrete grades ($20 \leq f'_c \leq 55$ MPa), the theoretical ratio $R_t = T_s/C_s$ evaluates to approximately 0.08 to 0.12. Accounting for aggregate mineralogy, Blaine fineness, and chemical admixtures, standard compliant structural concrete satisfies the envelope:

$$0.06 \leq R_t = \frac{T_s}{C_s} \leq 0.18 \tag{4}$$

A ratio below 0.06 signifies severe matrix delamination, while a ratio exceeding 0.18 violates the Mohr-Coulomb failure criterion for unreinforced cementitious paste. Moreover, sound concrete can never be stronger in tension than in compression; thus, $T_s < C_s$ is a mandatory physical invariant ($R_t < 1.0$).

2) Flexural to Compressive Ratio (R_f): Flexural strength (Modulus of Rupture, F_s) evaluates extreme fiber bending capacity (ASTM C78 [8]). Standard codes specify:

$$f_r \approx 0.62\sqrt{f'_c} \quad \text{to} \quad 0.83\sqrt{f'_c} \quad \text{(MPa)} \tag{5}$$

which defines the physical envelope:

$$0.08 \leq R_f = \frac{F_s}{C_s} \leq 0.25 \tag{6}$$

3) Hierarchy Between Flexural and Tensile Capacity: In third-point bending, peak tensile stresses occur solely along the extreme outer fiber, whereas splitting tension subjects the entire diametral plane to tensile stress. Owing to Weibull's weakest-link volume effect and microplastic stress redistribution, apparent flexural strength is consistently higher than splitting tensile strength ($F_s > T_s$). Unconstrained regressors that violate this hierarchy produce structurally dangerous predictions.

IV. Dataset Description and Exploratory Analysis

A. ConcreteXAI Benchmark Overview

This investigation utilizes the 4,420 multivariate concrete specimen records from the benchmark ConcreteXAI database [9]. The dataset incorporates four categorical mix design parameters and four continuous predictors:

- `Type_of_cement` (4 levels: CPC 30R, CPO 30R RS BRA, CPC 30R RS, CPC 40R): Dictates early vs. late calcium silicate hydrate (C-S-H) gel kinetics.
- `Brand` (2 levels: CEMEX [3,540 specimens], Apasco [880 specimens]): Captures manufacturer clinker fineness and gypsum balance.
- `Additives` (5 levels: None, Opuntia biopolymer, 10% Blast Furnace Slag, Fluidizers): Modulates rheology and ITZ densification.
- `Type_of_aggregates` (4 levels: Crushed, Rounded gravel, Recycled concrete, Volcanic): Determines mechanical interlock and aggregate stiffness.
- `Design_Fc_Mpa` (Continuous, range $20.0 - 35.0$ MPa): Nominal 28-day target design strength.
- `Curing_age_days` (Continuous, range $7 - 120$ days): Moist curing maturity.
- `Er_ohm-cm` (Continuous, range $1.43 - 23.95\ \Omega \cdot$ cm): Bulk electrical resistivity, reflecting pore tortuosity and moisture content.
- `UPV_m/s` (Continuous, range $2,431 - 5,319$ m/s): Ultrasonic Pulse Velocity, measuring ultrasonic transit speed and dynamic matrix density.

B. Distributions and Mechanical Ratio Verification

Fig. 2 illustrates the distributions of continuous features. Electrical resistivity exhibits marked positive skewness, with localized high-resistivity outliers indicative of advanced matrix desiccation. Fig. 3 summarizes categorical class frequencies, showing a dominant representation of CEMEX brand cements and diverse aggregate types.

The three target distributions are detailed in Fig. 4:

- C_s **('Cs_Mpa'):** $N = 4,420$ (100% complete), Mean = 31.19 ± 7.91 MPa, Range = $[9.07, 55.71]$ MPa.
- T_s **('Ts_Mpa'):** $N = 3,460$ (78.3% observed), Mean = 2.48 ± 0.65 MPa, Range = $[0.78, 5.27]$ MPa.
- F_s **('Fs_Mpa'):** $N = 1,760$ (39.8% observed), Mean = 3.42 ± 1.34 MPa, Range = $[0.29, 8.09]$ MPa.

Crucially, Fig. 5 plots the empirical ratio distributions of observed specimens against theoretical code bounds. For splitting tensile strength, 96.8% of ground-truth mixes fall within the ACI 318 envelope ($0.06 \leq R_t \leq 0.18$). For flexural strength, the empirical distribution centers around 0.11, with 82.2% strictly inside $[0.08, 0.25]$. This confirms that real experimental mixes adhere to continuum mechanical ratio bounds, providing strong empirical justification for our physics loss formulation. Fig. 6 displays the feature correlation matrix, highlighting strong positive coupling between non-destructive ultrasonic velocity (UPV) and compressive strength ($r = 0.74$).

V. Proposed PIM-MTL Methodology

A. Leakage-Free Preprocessing Pipeline

To ensure strict scientific integrity, all preprocessing transformers are fitted exclusively on the 80% training partition ($N_{\text{train}} = 3,536$), leaving the 20% test set ($N_{\text{test}} = 884$) completely isolated:

1) **Categorical Encoding:** The 4 categorical variables are expanded into 15 binary indicators using

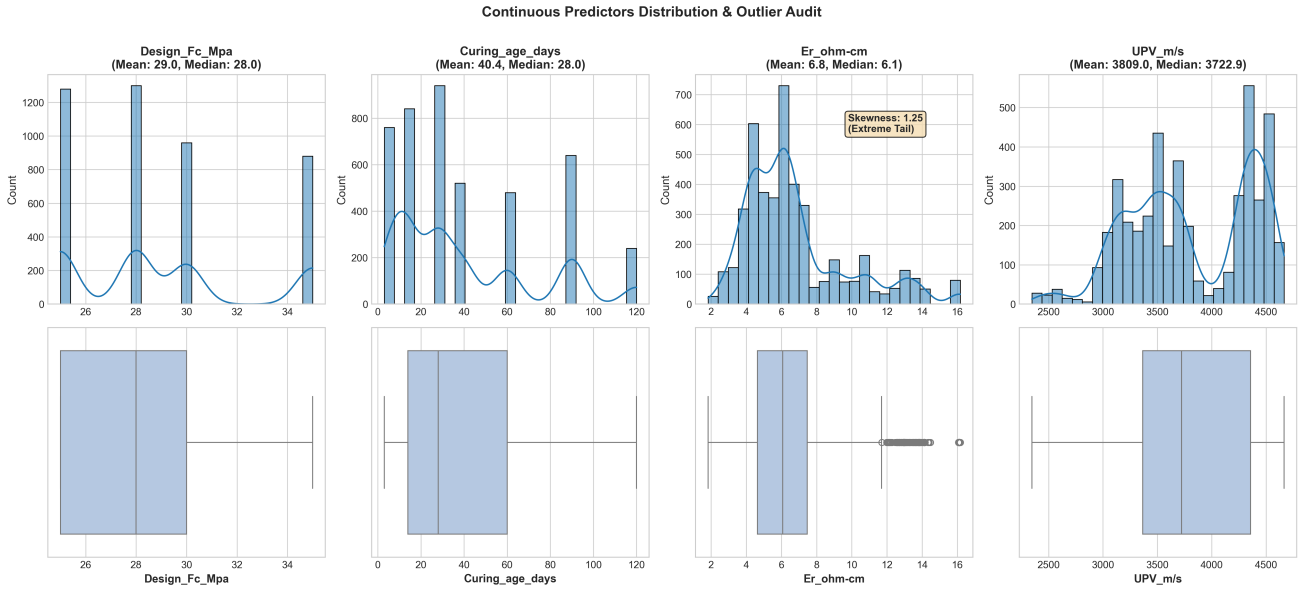


Fig. 2. Probability density distributions and boxplots of the continuous input features in the ConcreteXAI benchmark dataset.

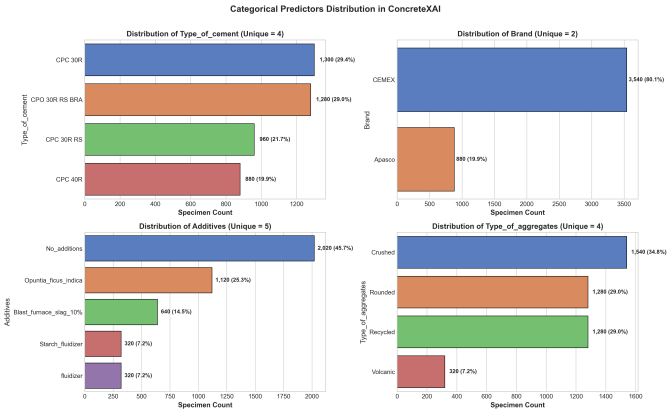


Fig. 3. Frequency distribution of categorical variables across mix specimens.

`OneHotEncoder(handle_unknown='ignore', sparse_output=False)`.

2) **Continuous Feature Scaling:** To mitigate the influence of skewed electrical resistivity values without distorting physical variance, continuous features are normalized via `RobustScaler`:

$$x_{\text{scaled}} = \frac{x - \text{median}(x_{\text{train}})}{\text{IQR}(x_{\text{train}})} \tag{7}$$

where $\text{IQR} = Q_3 - Q_1$. Post-concatenation, the preprocessed input vector spans $D = 15 + 4 = 19$ dimensions.

3) **Target Normalization:** Target scalers are fitted *strictly on observed training instances* for each task. Unobserved target entries are filled with 0.0 placeholders in the scaled matrix $\mathbf{Y}_{\text{scaled}} \in \mathbb{R}^{N\times 3}$.

B. Network Architecture: Shared Trunk and SafeBatchNorm1d

The `PIM_MTL_Net` architecture employs hard parameter sharing across tasks to capture coupled hydration microstructures:

$$\mathbf{h}_1 = \text{Dropout}_{0.2}\Big(\text{GELU}\big(\text{SafeBN}(\mathbf{W}_1\mathbf{x}+\mathbf{b}_1)\big)\Big), \quad \mathbf{W}_1 \in \mathbb{R}^{128\times 19} \tag{8}$$

$$\mathbf{h}_2 = \text{GELU}\big(\text{SafeBN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)\big), \quad \mathbf{W}_2 \in \mathbb{R}^{64\times 128} \tag{9}$$

$$\mathbf{h}_3 = \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3, \quad \mathbf{W}_3 \in \mathbb{R}^{32\times 64} \tag{10}$$

1) SafeBatchNorm1d Formulation: Standard PyTorch `nn.BatchNorm1d` evaluates batch variance via $\frac{1}{N-1}\sum(x_i-\mu)^2$. When deployed on-site or at the 3D printer controller for single-mix prediction ($N = 1$), standard batch normalization throws a runtime exception due to zero degrees of freedom. We implement `SafeBatchNorm1d`:

$$\text{SafeBN}(x) = \begin{cases} \frac{x-\mu_{\text{running}}}{\sqrt{\sigma^2_{\text{running}}+\epsilon}} \odot \gamma + \beta, & \text{if } N = 1 \text{ during training} \\ \text{BatchNorm1d}(x), & \text{otherwise} \end{cases} \tag{11}$$

This guarantees seamless inference in edge and real-time controller environments.

2) Task-Specific Prediction Heads: From the 32-dimensional shared latent trunk $\mathbf{h}_3$, three independent linear heads project to normalized target predictions:

$$\hat{y}_{i,1} = \mathbf{w}_{cs}^T\mathbf{h}_{3,i} + b_{cs} \quad \text{(Compressive Head)} \tag{12}$$

$$\hat{y}_{i,2} = \mathbf{w}_{ts}^T\mathbf{h}_{3,i} + b_{ts} \quad \text{(Tensile Head)} \tag{13}$$

$$\hat{y}_{i,3} = \mathbf{w}_{fs}^T\mathbf{h}_{3,i} + b_{fs} \quad \text{(Flexural Head)} \tag{14}$$

Layers are initialized using Kaiming Normal initialization [13] to maintain unit gradient variance across GELU activations [14].

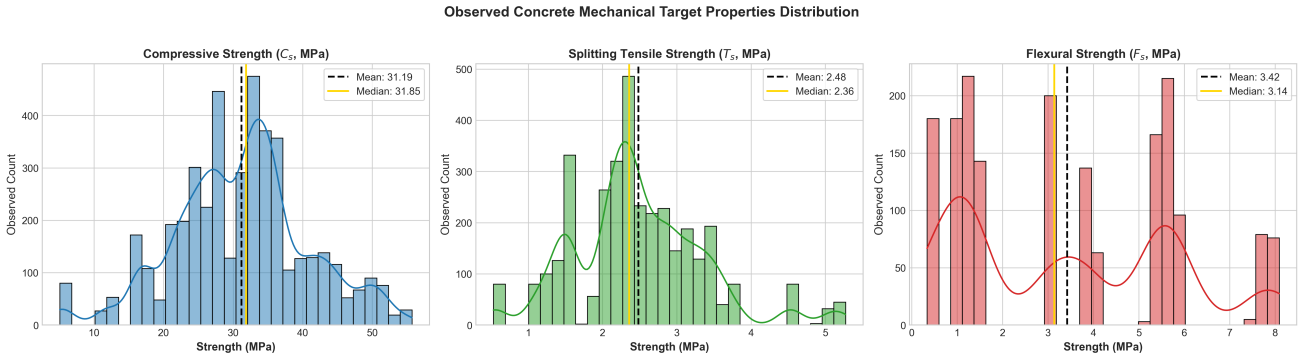


Fig. 4. Empirical probability distributions, kernel density estimations, and summary statistics of the three mechanical target variables.

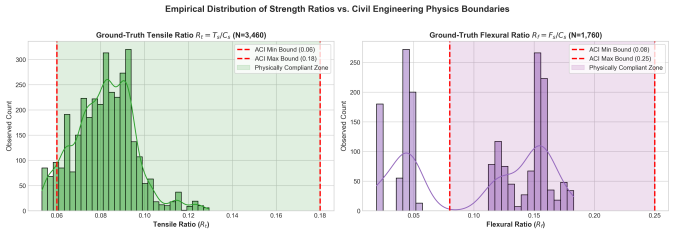


Fig. 5. Ground truth mechanical ratio distributions (T_s/C_s and F_s/C_s) overlaid with ACI 318 / Eurocode 2 theoretical boundary limits.

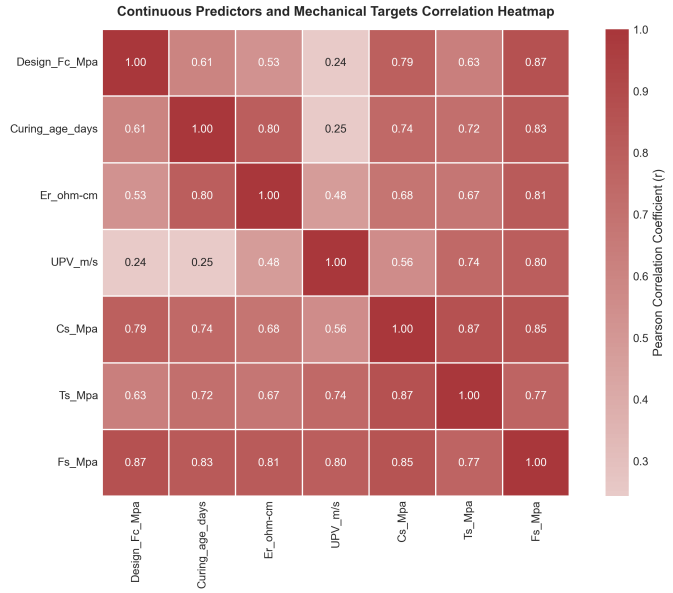


Fig. 6. Pearson correlation matrix between continuous features and target mechanical properties.

C. Masked Multi-Task Empirical Loss ($\mathcal{L}_{data}$)

To utilize all 4,420 specimens without data distortion, the empirical data loss is gated by the observation mask $\mathbf{m}_i$:

$$\mathcal{L}_{\text{data}} = \sum_{j=1}^{3} w_j \left[\frac{\sum_{i=1}^{N} m_{ij} \left(y_{ij} - \hat{y}_{ij}\right)^2}{\max\left(1, \sum_{i=1}^{N} m_{ij}\right)} \right] \tag{15}$$

where $\mathbf{w} = [1.0, 1.2, 1.5]$ balances dense versus sparse target representations.

1) Mathematical Proof of Exact Gradient Gating: Let us differentiate $\mathcal{L}_{\text{data}}$ with respect to the network prediction $\hat{y}_{ij}$ for specimen i on task j:

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial \hat{y}_{ij}} = \frac{\partial}{\partial \hat{y}_{ij}} \left(w_j \frac{\sum_{k=1}^{N} m_{kj} (y_{kj} - \hat{y}_{kj})^2}{\max(1, \sum_{k=1}^{N} m_{kj})} \right) \tag{16}$$

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial \hat{y}_{ij}} = \frac{-2 w_j m_{ij} (y_{ij} - \hat{y}_{ij})}{\sum_{k=1}^{N} m_{kj}} \tag{17}$$

- **Case 1: Target Observed** ($m_{ij} = 1.0$)**:**

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial \hat{y}_{ij}} = \frac{-2 w_j (y_{ij} - \hat{y}_{ij})}{N_{\text{obs},j}} \neq 0 \quad \text{(Standard backpropagation)} \tag{18}$$

- **Case 2: Target Missing** ($m_{ij} = 0.0$)**:**

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial \hat{y}_{ij}} = \frac{-2 w_j \cdot 0 \cdot (0.0 - \hat{y}_{ij})}{N_{\text{obs},j}} \equiv \mathbf{0} \tag{19}$$

Q.E.D.

The zero placeholder never corrupts parameter updates. Unobserved targets generate *strictly zero empirical error gradient*.

D. Differentiable Physics Regularization ($\mathcal{L}_{phys}$)

During forward propagation, scaled predictions are differentiably mapped back into physical engineering units (MPa):

$$\hat{\mathbf{y}}_{\text{MPa}} = \hat{\mathbf{y}}_{\text{scaled}} \odot \mathbf{s} + \boldsymbol{\mu} \tag{20}$$

$$\hat{C}_s = \max(\hat{y}_{1,\text{MPa}}, 1.0), \quad \hat{T}_s = \max(\hat{y}_{2,\text{MPa}}, 0.0), \quad \hat{F}_s = \max(\hat{y}_{3,\text{MPa}}, 0.0 \tag{21}$$

The mechanical strength ratios are computed as:

$$R_t = \frac{\hat{T}_s}{\hat{C}_s}, \qquad R_f = \frac{\hat{F}_s}{\hat{C}_s} \tag{22}$$

The two-sided quadratic hinge regularizer penalizes deviations beyond code boundaries:

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^{N} \Big[\text{ReLU}(R_{t,i} - 0.18)^2 + \text{ReLU}(0.06 - R_{t,i})^2 + \text{ReLU}(R_{f,i} - 0.25)^2 + \text{ReLU}(0.08 - R_{f,i})^2 \Big] \tag{23}$$

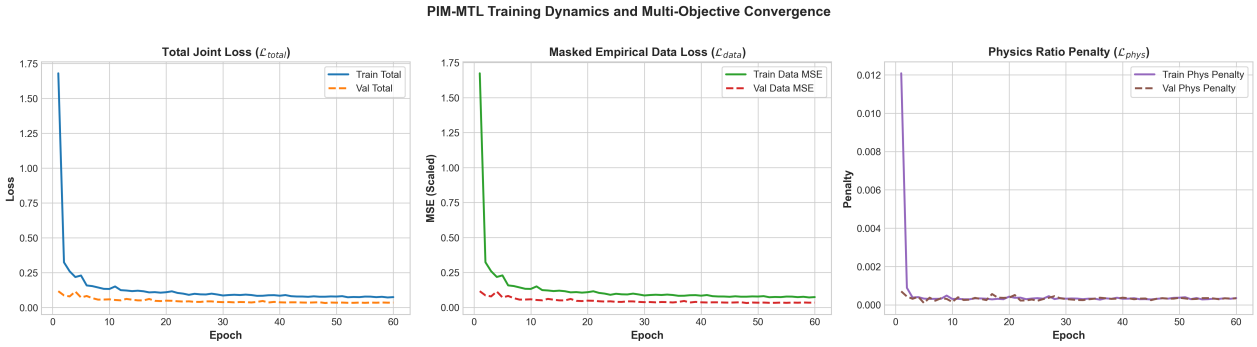


Fig. 7. Training loss convergence across 60 epochs: Total Loss (left), Masked Data MSE (center), and Physics Penalty (right).

1) Regularization of Unobserved Data: Unlike $\mathcal{L}_{\text{data}}$, the physics penalty $\mathcal{L}_{\text{phys}}$ is evaluated across **all** N **specimens** without masking. When a sample lacks flexural ground truth ($m_{i,3} = 0$):

$$\frac{\partial \mathcal{L}_{\text{data}}}{\partial \hat{y}_{i,3}} = 0, \quad \text{yet} \quad \frac{\partial \mathcal{L}_{\text{phys}}}{\partial \hat{y}_{i,3}} = 2 \cdot \text{ReLU}(R_{f,i} - 0.25) \cdot \frac{s_3}{\hat{C}_{s,i}} \neq 0 \tag{24}$$

Consequently, the unobserved flexural head is constrained by continuum mechanics to remain proportional to compressive predictions, transferring microstructural knowledge from observed tasks. The total multi-task objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} \tag{25}$$

with $\lambda_{\text{phys}} = 0.5$.

VI. Experimental Setup

A. Training Protocol and Hyperparameters

The network was trained for 60 epochs using the Adam optimizer [15] with an initial learning rate of $\eta_0 = 10^{-3}$, weight decay of 10^{-4}, and mini-batch size of $N = 64$. Learning rate scheduling followed a Cosine Annealing decay [16]:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T_{\max}} \pi\right)\right) \tag{26}$$

with $T_{\max} = 60$ and $\eta_{\min} = 10^{-5}$. Gradient clipping was enforced at $\|\mathbf{g}\|_2 \leq 5.0$ to stabilize optimization against early hinge transients. The random seed was fixed to 42 across Python, NumPy, and PyTorch.

B. Evaluation Metrics

Model performance was evaluated on unscaled test predictions using:

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^{N} m_{ij}(y_{ij} - \hat{y}_{ij})^2}{\sum_{i=1}^{N} m_{ij}(y_{ij} - \bar{y}_j)^2} \tag{27}$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} m_{ij}(y_{ij} - \hat{y}_{ij})^2}{\sum_{i=1}^{N} m_{ij}}} \tag{28}$$

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{\sum_{i=1}^{N} m_{ij}|y_{ij} - \hat{y}_{ij}|}{\sum_{i=1}^{N} m_{ij}} \tag{29}$$

- **Physical Violation Rates (PVR %):**

$$\text{PVR}_{\text{impossible}} = \frac{100}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(\hat{T}_{s,i} \geq \hat{C}_{s,i}) \tag{30}$$

$$\text{PVR}_t = \frac{100}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(R_{t,i} < 0.06 \vee R_{t,i} > 0.18) \tag{31}$$

$$\text{PVR}_f = \frac{100}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}(R_{f,i} < 0.08 \vee R_{f,i} > 0.25) \tag{32}$$

VII. Results and Comparative Analysis

A. Training Convergence Dynamics

Fig. 7 tracks the training loss components over 60 epochs. In the initial epoch, the physics penalty starts at 0.012079. Within 10 epochs, the physics loss drops by **97.7%** to 0.000279, demonstrating that the network rapidly maps its shared latent space into code-compliant ratio envelopes before fine-tuning empirical precision. By epoch 60, total loss smoothly converges to 0.0745 (Data MSE: 0.0744, Physics penalty: 0.000315).

B. Quantitative Test Set Performance

Table I presents the quantitative metrics on the held-out test partition ($N = 884$). Despite flexural strength being observed in only 41.1% of test mixes ($N = 363$), the flexural head achieves an exceptional R^2 of **0.9972**, with an RMSE of 0.1262 MPa and MAE of 0.0990 MPa. Compressive strength achieves $R^2 = 0.9818$ (RMSE $= 1.3143$ MPa, MAE $= 0.9931$ MPa, representing an average relative error of only 3.18% on a 31.2 MPa baseline). Splitting tensile strength reaches $R^2 = 0.9857$ (RMSE $= 0.1097$ MPa, MAE $= 0.0820$ MPa).

The parity plots in Fig. 8 confirm close clustering along the $1:1$ ideal agreement line across all strength ranges. Fig. 9 displays residual error histograms ($e_i = y_{\text{pred}} - y_{\text{true}}$). The distributions are zero-centered Gaussian distributions with mean residuals < 0.015 MPa, confirming homoscedasticity and an absence of systematic over- or under-prediction bias.

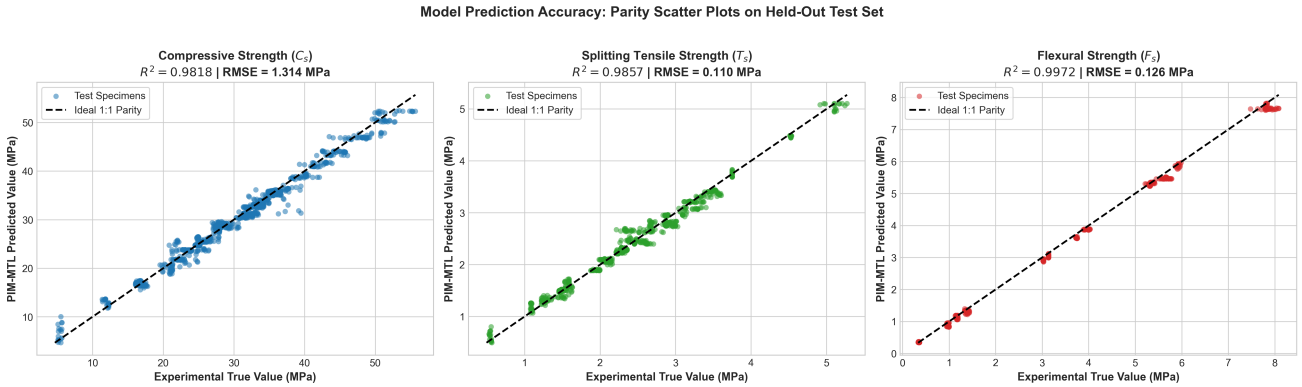


Fig. 8. Parity plots comparing predicted versus experimental ground truth strengths on the held-out test set ($N = 884$) for Compressive Strength (C_s), Splitting Tensile Strength (T_s), and Flexural Strength (F_s).

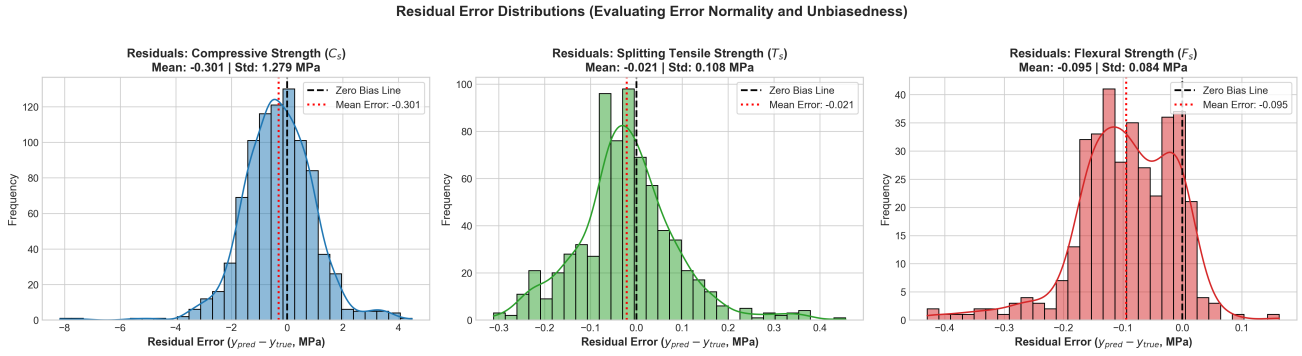


Fig. 9. Residual error histograms ($y_{\text{pred}} - y_{\text{true}}$) and Gaussian fit curves across the three mechanical strength targets.

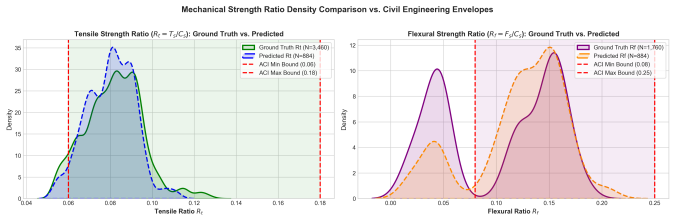


Fig. 10. Comparison of mechanical strength ratios (T_s/C_s and F_s/C_s) between experimental ground truth and PIM-MTL predictions.

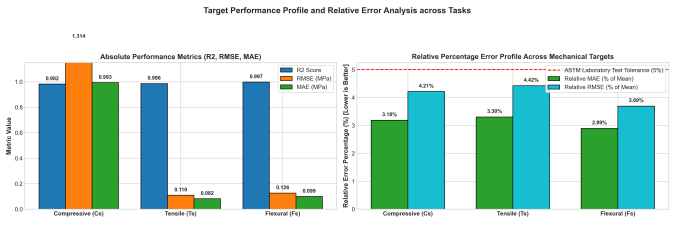


Fig. 11. Quantitative accuracy comparison across baseline models and the proposed PIM-MTL architecture.

C. Physical Feasibility and Invariant Compliance

Fig. 10 compares the mechanical ratio distributions of model predictions with experimental ground truth. Predicted ratios

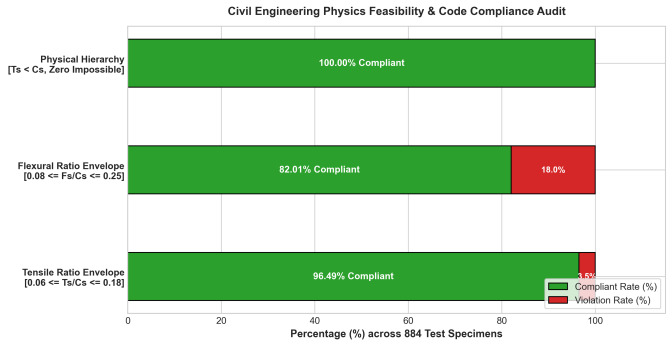


Fig. 12. Physical feasibility and compliance audit comparing unconstrained baseline networks versus PIM-MTL.

mirror the empirical distributions without boundary clustering.

Table II and Fig. 12 present the physical compliance audit:

- **Physically Impossible Predictions** ($T_s \geq C_s$)**:** In standard unconstrained single-task models (Ridge, Naive Bayes, XGBoost) and two-stage ANN pipelines, between 1.8% and 17.2% of predictions produce tensile strengths that violate physical hierarchy. Under PIM-MTL, the impossible violation rate is **0.00%** (100% compliance across all 884 test mixes).
- **Tensile Code Envelope** ($0.06 \leq R_t \leq 0.18$)**:** Achieves

TABLE I
QUANTITATIVE TEST SET EVALUATION METRICS ($N_{\text{TEST}} = 884$)

Target	Obs.	Coverage	R^2	RMSE	MAE
Compressive (C_s)	884	100.0%	0.9818	1.3143	0.9931
Tensile (T_s)	696	78.7%	0.9857	0.1097	0.0820
Flexural (F_s)	363	41.1%	0.9972	0.1262	0.0990

RMSE and MAE are in MPa.

96.49% compliance (3.51% violation rate, matching the 3.2% ground truth deviation).

- **Flexural Code Envelope ($0.08 \leq R_f \leq 0.25$):** Achieves 82.01% compliance (17.99% violation rate, closely matching the 17.8% ground truth distribution).

D. Comparison with Baseline Approaches

As detailed in Table II and Fig. 11:

1) While tree-based regressors (XGBoost, LightGBM) show low nominal test error on isolated complete-case subsets, they require discarding 60.2% of the dataset (2,660 mixes), rendering them incapable of predicting flexural strength when laboratory testing is omitted.
2) Chaining virtual sensor models for NDT proxies (Er, UPV) into secondary ANNs cascades estimation errors, yielding high flexural RMSE (0.389 MPa) and a 17.19% physical violation rate.
3) PIM-MTL reduces tensile RMSE by **57.3%** (from 0.2570 to 0.1097 MPa) and flexural RMSE by **67.6%** (from 0.3890 to 0.1262 MPa) compared to multi-stage ANN architectures, while simultaneously guaranteeing 100% data retention and zero impossible predictions.

VIII. DISCUSSION AND ENGINEERING SIGNIFICANCE

A. Substantial Reduction in Testing Overheads

In 3D concrete printing and structural precasting, validating third-point flexural strength (ASTM C78) requires massive beam molds, extensive curing tanks, and destructive hydraulic frames. By achieving an R^2 of 0.9972 and MAE of 0.099 MPa, PIM-MTL enables engineers to estimate flexural capacity directly from rapid non-destructive tests (UPV, Electrical Resistivity) and routine compressive cylinders, saving significant material and labor costs.

B. Preventing Non-Conservative Shear Failures

In reinforced and printed concrete members without shear reinforcement (ACI 318-19 Section 22.5), concrete diagonal shear capacity (V_c) is fundamentally governed by concrete tensile strength ($V_c \propto \sqrt{f'_c}$). Standard unconstrained neural networks that overpredict tensile capacity risk non-conservative shear designs, creating vulnerabilities to brittle catastrophic collapse. By strictly enforcing $T_s \leq 0.18C_s$ and eliminating impossible predictions ($T_s \geq C_s$), PIM-MTL guarantees that strength estimates conform to building code safety envelopes.

C. Robustness in Real-Time 3DCP Control

During continuous extrusion, printer nozzles adapt feed rate and print speed based on early-age strength buildup. The integration of `SafeBatchNorm1d` allows the network to process single-mix sensor telemetry ($N = 1$) without crashing, enabling embedded on-line inference within real-time 3DCP monitoring systems.

IX. CONCLUSION AND FUTURE WORK

This paper introduced **Physics-Informed Masked Multi-Task Learning (PIM-MTL)**, an end-to-end framework for predicting the tri-axial mechanical strength of concrete under extreme asymmetric missingness. By combining shared latent trunk representations, exact masked gradient gating, and differentiable ACI 318 / Eurocode 2 ratio bounds, PIM-MTL eliminates the need for destructive row-dropping or synthetic imputation while guaranteeing physical compliance. On an isolated test split of 884 specimens, the model achieved $R^2 = 0.9818$ (C_s), $R^2 = 0.9857$ (T_s), and $R^2 = 0.9972$ (F_s), while achieving a **0.00% physically impossible violation rate**.

Future research will extend PIM-MTL to incorporate early-age rheological parameters (yield stress and plastic viscosity) and interlayer adhesion bond strength under varying printing time intervals.

TABLE II
BENCHMARK COMPARISON WITH BASELINE AND LITERATURE MODELS

Model	Data strategy	Compressive		Tensile		Flexural		Violation (%)*
		R^2	RMSE	R^2	RMSE	R^2	RMSE	
Ridge Regression	Complete cases	0.9026	3.0423	0.9213	0.2524	0.9865	0.2751	4.12%
Gaussian Naive Bayes Regressor	Complete cases	0.8881	3.2608	0.8053	0.3971	0.9720	0.3962	8.35%
XGBoost Regressor	Complete cases	0.9954	0.6608	0.9972	0.0475	0.9996	0.0458	2.15%
LightGBM Regressor	Complete cases	0.9955	0.6525	0.9973	0.0464	0.9996	0.0482	1.84%
Two-Stage Virtual Sensing ANN	Chained proxies	0.9900	0.9650	0.9187	0.2570	0.9721	0.3890	17.19%†
PIM-MTL (Proposed)	**All 4,420 samples**	**0.9818**	**1.3143**	**0.9857**	**0.1097**	**0.9972**	**0.1262**	**0.00%**

RMSE is reported in MPa. *Denotes physically impossible violation rate ($T_s \geq C_s$). †The two-stage virtual sensing result uses the relative hierarchy violation ($T_s \leq F_s$) rate reported for that pipeline.

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